NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 10/01/2026
Time: 3 Hrs | GTM – 14 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	C	D	B	C	A	A	D	A	A
11	12	13	14	15	16	17	18	19	20
B	B	B	B	B	A	A	B	B	B
21	22	23	24	25					
60	400	4	4	0					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	A	B	D	C	C	B	A	C	A
36	37	38	39	40	41	42	43	44	45
A	B	C	A	B	A	B	D	A	C
46	47	48	49	50					
328	4	30	4	3					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	B	D	B	A	D	C	B	D	A
61	62	63	64	65	66	67	68	69	70
B	A	A	B	A	B	D	C	D	D
71	72	73	74	75					
50	4	8	9	2					

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HINTS
PHYSICS

1. $|\hat{e}_1 - \hat{e}_2| = \sqrt{1^2 + 1^2 - 2\times1\times1(\cos\theta)}$

$= \sqrt{2 - 2\cos\theta} = \sqrt{2(1-\cos\theta)}$

$= \sqrt{2.2\sin^2\left(\frac{\theta}{2}\right)} = 2\sin\left(\frac{\theta}{2}\right)$

2.

$\left[\frac{d}{dt}\int F.ds)\right] = [A(F.p]$

$\left[\frac{Fs}{t}\right] = [AFp] \Rightarrow [A] = \left[\frac{s}{pt}\right] = \left[\frac{L}{MLT^{-1}T}\right] = [M^{-1}L^0T^0]$

3. $s = \frac{1}{2}ft^2 \Rightarrow t = \sqrt{\frac{2s}{f}}$

$v^2 = u^2 + 2as \Rightarrow v = \sqrt{2fs}$

$s_2 = \left(\sqrt{2fst}\right)$

$v^2 = u^2 + 2as \Rightarrow 0^2\left(\sqrt{2fs}\right)^2 - 2\left(\frac{f}{2}\right)s_3$

$\Rightarrow s_2 = 2s$

$s_2 = 15s - s - 2s = 12s\sqrt{2fst}$

$\Rightarrow s = \frac{1}{72}ft^2$

4. Drift $= V_R\left(\frac{d}{V_{MR}}\right) \Rightarrow 300 = 3\left(\frac{500}{V_{MR}}\right)$

$\Rightarrow V_{MR} = 5$ mph

5. $T = \frac{2m_1m_2g_{eff}}{m_1+m_2} = \frac{2(3)(1.5)}{4.5}\left(g + \frac{g}{10}\right) = \frac{11g}{5} = 22N$

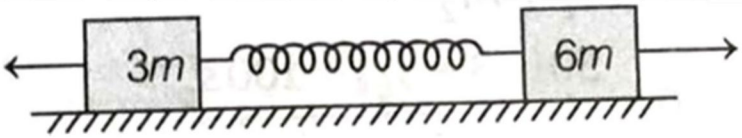
Reading of spring balance $= \frac{44}{g} = 4.4$ kg

6. $F_{ext} = ma_{CM} \Rightarrow F - \frac{F}{2} = (6m+3m)a_{CM}$

$\Rightarrow a_{CM} = \frac{F}{18cm}$

In CM reference frame, each block experiences pseudo force. $\frac{2F}{3} = \frac{F}{2} + 3m\left(\frac{F}{18m}\right)$

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$$F-6m\left(\frac{F}{18m}\right)=\frac{2F}{3}$$

Using work-energy theorem in CM reference frame,

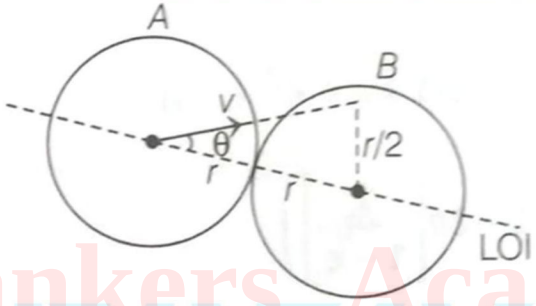
$$W_{ext}+W_{spring}=\Delta K$$

$$\Rightarrow\left(\frac{2F}{3}\right)x_1+\left(\frac{2F}{3}\right)x_2+\frac{1}{2}K[0^2-(x_1+x_2)^2]=0-0$$

$$\Rightarrow x_1+x_2=\frac{4F}{3K}$$

So, maximum elongation will be $\frac{4F}{3K}$

7.



$$\sin\theta=\frac{r/2}{2r}=\frac{1}{4}$$

Velocity of A along the line of impact before collision

$$=v\cos\theta=v\sqrt{1-\left(\frac{1}{4}\right)^2}=\frac{\sqrt{15}v}{4}$$

Along the LOI, velocities of two balls will get exchanged. So, velocity of ball B after collision $=\frac{\sqrt{15}}{4}v$

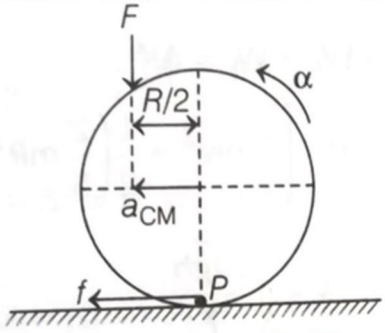
8. $\sum\tau=0$

$$\Rightarrow Mg\left(\frac{L}{2}-\sin 30^\circ\right)-mg\left(\frac{L}{2}\sin 60^\circ\right)=0$$

$$\Rightarrow\frac{M}{m}=\sqrt{3}$$

9. $\sum F_x=ma_x\Rightarrow f=ma_{CM}$

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$$\sum \tau = I_{CM}\alpha \Rightarrow -fR + F\left(\frac{R}{2}\right) = \left(\frac{mR^2}{2}\right)\alpha \quad \ldots(ii)$$

$$a_p = 0 \Rightarrow a_{CM} - R\alpha = 0 \quad \ldots(iii)$$

Solving Eqs. (i), (ii) and (iii), we get

$$f = \frac{F}{3}$$

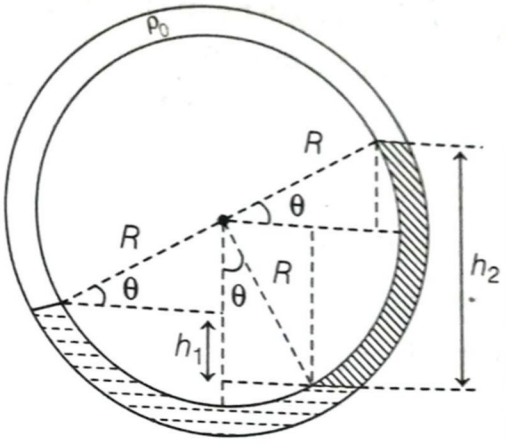
10. $r_p = a(1-e);\ e_a = a(1+e)$

$$mv_p r_p = mv_a r_a \Rightarrow \frac{v_p}{v_a} = \frac{1+e}{1-e}$$

11. $$k_{eq} = k_1 + k_2 = \frac{(2A)Y_{eq}}{L} = \frac{AY_1}{L} + \frac{AY_2}{L}$$

$$\Rightarrow Y_{eq} = \frac{Y_1 + Y_2}{2}$$

12.



Pressure at interface = $\rho_0 + \rho g h_1 = \rho_0 + \sigma g h_2$

$$\Rightarrow \rho g(R\cos\theta - R\sin\theta) = \sigma g(R\cos\theta + R\sin\theta)$$

$$\Rightarrow \rho\cos\theta - \rho\sin\theta = \sigma\cos\theta + \sigma\sin\theta$$

$$\Rightarrow \tan\theta = \frac{\rho - \sigma}{\rho + \sigma}$$

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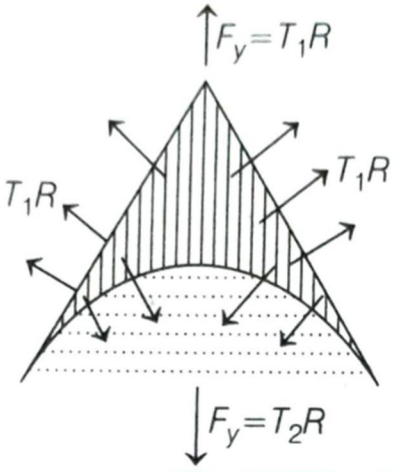
13. $6\pi\eta RV = Mg\left[1-\frac{\rho_L}{\rho_S}\right]$

Clearly $R\times V$ Constant

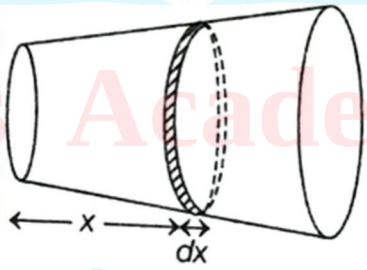
14. $F_y = T\times$ Projected length

Net force = Netforce $= T_1R - T_2R$

$= (T_1 - T_2)R$ in + y-direction



15. Let us taken an element of width dx at a distance x from one end as shown in figure.



Thermal resistance of this element,

$$dR = \frac{dx}{K\pi\left(\frac{d_0}{2}(1+ax)\right)^2} = \frac{4dx}{K\pi d_0^2(1+ax)^2}$$

For finding thermal resistance of rod,

$$\int dR = \frac{4}{K\pi d_0^2}\int_0^1 \frac{dx}{(1+ax)^2}$$

$$\Rightarrow R = \frac{4}{K\pi d_0^2}\frac{1}{a}\left[-\frac{1}{(1+ax)}\right]_0^{\ell} = \frac{4\ell}{K\pi d_0^2(a\ell+1)}$$

17. When $V = V_0$

$$p_1 = p_0\left[1+\left(\frac{2V_0}{V_0}\right)^2\right]^{-1} = \frac{p_0}{5}$$

When $V = 2V_0$,

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$$p_2 = p_0\left[1+\left(\frac{2V_0}{2V_0}\right)^2\right]^{-1} = \frac{p_0}{2}$$

$$T_2 - T_1 = \frac{p_2V_2 - p_1V_1}{nR} = \frac{\left(\frac{p_0}{2}\right)2V_0 - \left(\frac{p_0}{5}\right)V_0}{nR} = \frac{4}{5}\frac{p_0V_0}{nR}$$

18. $C = C_v + \dfrac{R}{1-n}$

$$\Rightarrow C = \frac{R}{\gamma-1} + \frac{R}{1-n} < 0$$

$$\Rightarrow \frac{(\gamma-n)R}{(\gamma-1)(1-n)} < 0$$

As, $\gamma > 1,\ \gamma > n > 1$

19. For superposition of x_1 and x_2,

$$A' = \sqrt{A^2 + A^2 + 2(A)(A)\cos\left(\frac{2\pi}{3}\right)} = A$$

$$\psi = \tan^{-1}\left[\frac{A\sin\left(\frac{2\pi}{3}\right)}{A + A\cos\left(\frac{2\pi}{3}\right)}\right] = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$(x_1 + x_2) + x_3 = 0$$

$$\Rightarrow A\sin\left(\omega t + \frac{\pi}{3}\right) + B\sin(\omega t + \phi) = 0$$

$$\Rightarrow B\sin(\omega t + \phi) = -A\sin\left(\omega t + \frac{\pi}{3}\right)$$

$$\Rightarrow B\sin(\omega t + \phi) = A\sin\left(\omega t + \frac{\pi}{3} + \pi\right)$$

$$\Rightarrow B\sin(\omega t + \phi) = A\sin\left(\omega t + \frac{4\pi}{3}\right)$$

$$\therefore B = A, \phi = \frac{4\pi}{3}$$

20. $x = f(x - vt) = \dfrac{6}{(x-2t)^2}$

$$v_p = \frac{\partial y}{\partial t} = -\frac{12}{(x-2t)^3}(-2) = \frac{24}{(x-2t)^3}$$

At $x = 2, t = 2$,

$$v_p = \frac{24}{[2-2(2)]^3} = -3\,m/s$$

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21. Let the liquid column in arm A be depressed by a distance y.

$p_1 = p_2$

$\Rightarrow \rho_2 g h_2 = \rho_1 g (2y)$

$\Rightarrow \left(\frac{\rho_1}{2}\right) h_2 = \rho_1 (2y)$

$\Rightarrow h_2 = 4y$

2nd overtone of A = 1st harmonic of B

$\Rightarrow \frac{5v}{4(70 - h_2 + y)} = \frac{v}{4(20 - y)}$

$\Rightarrow 6y - h_2 = 30$

$\Rightarrow 6y - 4y = 30$

$\Rightarrow y = 15 \text{ cm}$

$\therefore h_2 = 4y = 60 \text{ cm}$

22. $W_{BC} = -1830 \text{ J}, W_{CA} = 0$

For the cyclic process, $Q = \Delta U + W$

$\Rightarrow -1000 = 0 + (W_{AB} - 1830 + 0)$

$\Rightarrow W_{AB} = 830 \text{ J}$

$W_{AB} = nR\Delta T$

$\Rightarrow 830 = 1 \times 8.3 \times (T_B - 300)$

$\Rightarrow T_B = 400 \text{ K}$

23. $AF = AB + EF - CD = 2\ell_1 - \ell_2$

At new temperature,

$AF = 2\ell_1(1 + \alpha_1 \Delta T) - \ell_2(1 + \alpha_2 \Delta T)$

Distance AF remains same at all temperatures.

$2\ell_1 - \ell_2 = 2\ell_1(1 + \alpha_1 \Delta T) - \ell_2(1 + \alpha_2 \Delta T)$

$\Rightarrow \frac{\ell_2}{\ell_1} = \frac{2\alpha_1}{\alpha_2} = \frac{2(2 \times 10^{-5})}{1 \times 10^{-5}} = 4$

24. $A_1 v_1 = A_2 v_2$

$\pi 2^2 v_1 = \pi \left(\frac{1}{2}\right)^2 v_2$

$4 \times \frac{1}{4} = \frac{1}{4} v_2$

$R = v_2 \sqrt{\frac{2h}{g}} = 4\sqrt{\frac{2 \times 5}{10}} = 4m$

25. Let coordinates of point C be (x, y)

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$$\tan 37^\circ = \frac{y}{x-\frac{10}{3}} \Rightarrow y = \frac{3x}{4} - \frac{5}{2}$$

Equation of trajectory,

$$y = x\tan\alpha - \frac{gx^2}{2(u\cos\alpha)^2}$$

$$\Rightarrow \frac{3x}{4} - \frac{5}{2} = x\left(\frac{1}{2}\right) - \frac{10x^2}{2\left[5\sqrt{5}\left(\frac{2}{\sqrt{5}}\right)\right]^2}$$

$$\Rightarrow x^2 + 5x - 50 = 0 \Rightarrow x = 5m$$

$$y = \frac{3(5)}{4} - \frac{5}{2} = 1.25 \text{ m}$$

$$v_y^2 = u_y^2 + 2a_y s_y$$

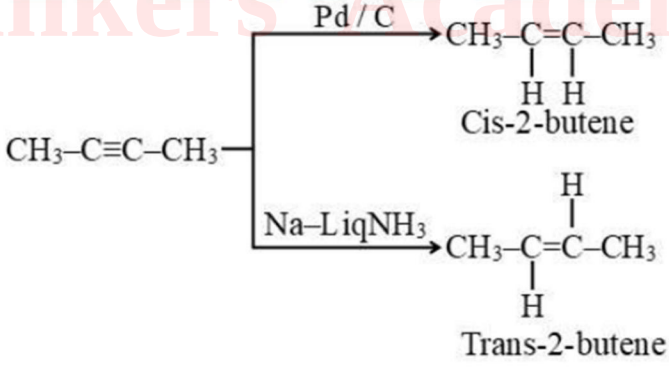
$$\Rightarrow v_y^2 = \left[5\sqrt{5}\left(\frac{1}{\sqrt{5}}\right)\right]^2 + 2(-10)(1.25)$$

$$\Rightarrow v_y = 0$$

So, the particle strikes the inclined plane at point C horizontaly

CHEMISTRY

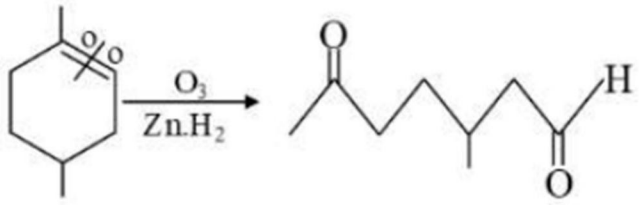
39.



	Cis-2-butene (A)	Trans-2 butene (B)
(i) Solubility	High	Low (correct)
(ii) Boiling point	High	Low (correct)
Melting point	Low	High (incorrect)
(iii) Polarity	High	Low (correct)
Dipole moment	High	Zero (incorrect)
Addition of Br_2	High	Low(incorrect)

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40.



3-Methyl-6-ketoheptanal

41.

(b) $CH_3 - COOK \xrightarrow{\text{electrolysis}} CH_3 - CH_3 + CO_2$

(c) $(CH_3)_2CHCOOH \xrightarrow[\text{reduction}]{HI/P} (CH_3)_2CH-CH_3$

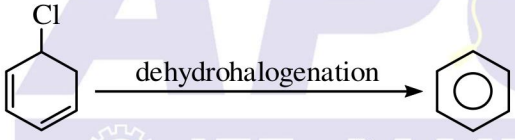
(d) $CH_3 - CH_2 - Br \xrightarrow[EtoH]{Zn-Cu\ couple} CH_3 - CH_3$

42. Na reacts with C, N and S to form NaCNS (sodium thiocyanate)

43.

	List-I (Purification technique)		List-II (Mixture of organic compounds)
(A)	Distillation (simple)	(III)	Chloroform + Aniline
(B)	Fractional distillation	(I)	Diesel + Petrol
(C)	Distillation under reduced pressure	(IV)	Glycerol + Spent-lye
(D)	Steam distillation	(II)	Aniline + Water

44.



Benzene (aromatic)
Highly stable

When a stable product is formed that means dehydrohalogenation becomes very easy.

45. Rate of Hydration $\propto$ Electron density on Alkene.

50. $x = 2.18\times10^6 \times \frac{1}{2}, \quad y = 2.18\times10^6 \times \frac{2}{6}$

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$$\frac{2x}{y}=\frac{2\times 2.18\times 10^6\times\frac{1}{2}}{2.18\times 10^6\times\frac{2}{6}}=3$$

MATHEMATICS

51. $\Delta=\begin{vmatrix}3 & 1 & \beta\\ 2 & \alpha & -1\\ 1 & 2 & 1\end{vmatrix}=0 \Rightarrow 3\alpha+4\beta-\alpha\beta+3=0$

$\Rightarrow \Delta_3=\begin{vmatrix}3 & 1 & \beta\\ 2 & \alpha & -1\\ 1 & 2 & 1\end{vmatrix}=0 \Rightarrow 9\alpha+19=0$

$\Rightarrow \alpha=\frac{-19}{9},\ \beta=\frac{6}{11} \Rightarrow 22\beta-9\alpha=31$

52. Slope of given line = $-\frac{3}{2}$

First of all we try to locate points on the curve at which tangents are parallel to given line

On differentiating $3x^2-4y^2=72$ w.r.t. x

$\frac{dy}{dx}=\frac{3x}{4y}=-\frac{3}{2}$ (Given) $\frac{x}{y}=-2$

Put $x=-2y$ in $3x^2-4y^2=72$

We get (–6, 3) and (6, –3) and their distances from line are $\frac{11}{\sqrt{13}}$ and $\frac{13}{\sqrt{13}}$ respectively

So (–6, 3) is nearest

So $x_0=-6 \Rightarrow y_0=3$ and $(x_0+y_0)=-3$

53. Given limit = $0+(2^2-1)+(3^2-1)+\ldots..+(10^2-1)=\sum_{n=2}^{10}(n^2-1)$

$=\frac{10\times 11\times 21}{6}-1-9=375$

54. $\frac{f(3x+3y)}{\sin(2x+3y)}=\frac{f(3x-3y)}{\sin(2x-2y)}\ \forall\ x, y\in R$

$\Rightarrow f(x)=k\sin\left(\frac{2x}{3}\right)$

$\therefore f'(0)=\frac{2}{3}k=1 \Rightarrow k=\frac{3}{2}$

$\therefore f(x)=\frac{3}{2}\sin\left(\frac{2x}{3}\right) \Rightarrow f'\left(\frac{\pi}{2}\right)=\frac{3\sqrt{3}}{4}$

Equation of tangent is $y-\frac{3\sqrt{3}}{4}=\frac{1}{2}\left(x-\frac{\pi}{2}\right)$

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56.

$$\text{Here, } g(x)=\begin{cases} f(x) & , \text{ if } 0\le x<\frac{1}{2} \\ f\left(\frac{1}{2}\right) & , \text{ if } \frac{1}{2}\le x\le 1 \\ 3-x & , \text{ if } 1<x\le 2 \end{cases}$$

57. $V_1=(\vec{a}+\vec{b})\cdot((\vec{b}+\vec{c})\times(\vec{c}+\vec{a}))=2[\vec{a}\vec{b}\vec{c}]$

$$V_2=\frac{1}{6}(\vec{a}\times\vec{b})\cdot[(\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a})]$$

$$=\frac{1}{6}(\vec{a}\times\vec{b})[((\vec{b}\times\vec{c})\cdot a)\vec{c}-((\vec{b}\times\vec{c})\cdot\vec{c})\vec{a}] \quad =\frac{1}{6}[\vec{a}\vec{b}\vec{c}]^2$$

$$[\vec{a}\vec{b}\vec{c}]=\begin{vmatrix} -3 & 1 & 1 \\ 4 & 2 & 4 \\ 2 & 2 & 0 \end{vmatrix}=36$$

$\therefore V_1=72$ and $V_2=216$

59.

$$\text{Given, } \sin\left(\frac{\pi}{2n}\right)+\cos\left(\frac{\pi}{2n}\right)=\frac{\sqrt{n}}{2}$$

$$\Rightarrow \sin^2\left(\frac{\pi}{2n}\right)+\cos^2\left(\frac{\pi}{2n}\right)+2\sin\left(\frac{\pi}{2n}\right)\cos\left(\frac{\pi}{2n}\right)=\frac{n}{4}$$

$$\Rightarrow 1+\sin\left(\frac{\pi}{n}\right)=\frac{n}{4} \Rightarrow \sin\left(\frac{\pi}{n}\right)=\frac{n-4}{4}$$

For n = 2, the given equation is not satisfied considering $n > 1$ and $n \neq 2$, $0<\sin\left(\frac{\pi}{n}\right)<1$

$$\Rightarrow 0<\frac{n-4}{4}<1 \Rightarrow 4<n<8$$

60.

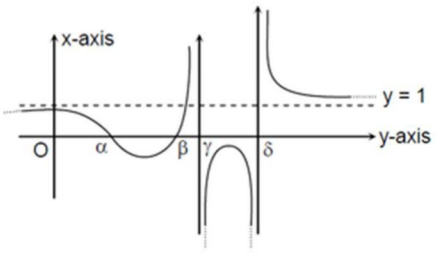
$$g(f(x))=\tan\left(x-\frac{\pi}{4}\right)=\frac{\tan x-1}{\tan x+1}$$

$$\Rightarrow g(x)=\frac{x-1}{x+1}$$

$$f(g(x))=\tan\left(\frac{x-1}{x+1}\right)$$

61.

$$\text{Graph of } f(x)=\frac{(x-\alpha)(x-\beta)}{(x-\gamma)(x-\delta)}$$



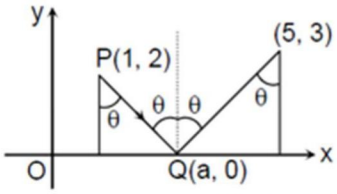
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62.

Image of P in x-axis is (1, –2)

$\therefore$ Equation of reflected ray is $(y-3)=\frac{5}{4}(x-5)$

$\Rightarrow 5x-4y=13$



63.

$\because$ $2\cos 2x-1=2(\cos^2 x-\sin^2 x)-(\sin^2 x+\cos^2 x)=\cos^2 x-3\sin^2 x$

The given equation reduces to $b\sin x=b+\sin x$, then

$\sin x=\frac{b}{b-1}$

$\Rightarrow -1\le \sin x\le 1$

$\Rightarrow -1\le \frac{b}{b-1}\le 1$

$\Rightarrow \frac{b}{b-1}\ge -1$

and $\frac{b}{b-1}\le 1$

$\Rightarrow b<\frac{1}{2}$

$\Rightarrow b\in\left(-\infty,\frac{1}{2}\right)$

64.

$f^{-1}(x)=x-1$

$\Rightarrow h^{-1}(x)=x^3+x$

$\therefore \frac{d}{dx}\left(h^{-1}(x)\right)=3x^2+1$

65. $\lim\limits_{x\to 2p^+}\left(\frac{1-\cos\left(x^2-4px+q^2+8q+16\right)}{\left(x^2-4px+q^2+8q+16\right)^2}\right)\left(\frac{\left(x^2-4px+q^2+8q+16\right)^2}{(x-2p)^2}\right)$

$\lim\limits_{h\to 0}\frac{1}{2}\left(\frac{(2p+h)^2-4p(2p+h)+q^2+82+16}{h^2}\right)=\frac{1}{2}$

Using L' Hospital's

66. $A(3,\alpha,3)$ & $B(-3,-7,\beta)$

$\overrightarrow{BA}=6\hat{i}+(\alpha+7)j+(3-\beta)\hat{k}$

$\vec{p}\times\vec{q}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&-1&1\\-3&2&4\end{vmatrix}$ $\Rightarrow \frac{|\overrightarrow{BA}\cdot(\vec{p}\times\vec{q})|}{|\vec{p}\times\vec{q}|}=3\sqrt{30}$

$\Rightarrow 36+15(\alpha+7)-3(3-\beta)=(3\sqrt{30})^2 \Rightarrow 36+15\alpha+105-9+3\beta=270$

$\Rightarrow 15\alpha+3\beta=138 \Rightarrow 5\alpha+\beta=46$

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70.

Using $\tan\theta - \tan\frac{\theta}{2} = \tan\frac{\theta}{2} + \tan\theta\tan^2\frac{\theta}{2}$

71. $\vec{r} = k(\vec{b} + \vec{c})$ and $\vec{r}\cdot\vec{a} = 3$

$\vec{r}\cdot\vec{a} = k(\vec{b}\cdot\vec{a} + \vec{c}\cdot\vec{a})$

$3 = k(2 + 6 - 15 + 3 - 2 + 3\lambda)$

$3 = k(3\lambda - 6) \Rightarrow k(\lambda - 2) = 1 \to 1$ and $|r| = k\sqrt{25 + 4 + (\lambda - 5)^2} = 1$

$\Rightarrow = \frac{1}{\lambda - 2}\sqrt{\lambda^2 - 10\lambda + 54} = 1 \quad \Rightarrow = \lambda^2 - 10\lambda + 54 = \lambda^2 + 4 - 4\lambda$

$\Rightarrow 6\lambda = 50$

72.

$\sin^2 x[\sin^2 x + \sin x + 2] = 0$

$\Rightarrow \sin x = 0$, where $x = 0, \pi, 2\pi, 3\pi$

75.

$$\text{Sum} = \sum_{n=1}^{n} \tan^{-1}\left[\frac{\sqrt{2n+2} - \sqrt{2n-2}}{2 + \sqrt{n^2 - 1}}\right] = \sum_{n=1}^{n} \tan^{-1}\left[\frac{\sqrt{\frac{n+1}{2}} - \sqrt{\frac{n-1}{2}}}{1 + \sqrt{\frac{n+1}{2}}\cdot\sqrt{\frac{n-1}{2}}}\right] \text{ (divide by 2)}$$

$$= \sum_{n=1}^{n}\left(\tan^{-1}\sqrt{\frac{n+1}{2}} - \tan^{-1}\sqrt{\frac{n-1}{2}}\right) = \tan^{-1}\left(\sqrt{\frac{n+1}{2}}\right) + \tan^{-1}\sqrt{\frac{n-1}{2}} - \tan^{-1}\frac{1}{\sqrt{2}}$$

$$\lim_{n\to\infty} \text{sum} = \pi - \tan^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{2} + \cot^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{2} + \tan^{-1}\left(\sqrt{2}\right)$$

$\therefore k = \sqrt{2} \Rightarrow k^2 = 2$.

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